

SOCIOL 723: Social Statistics II

Week 1 (Thursday), OLS in Matrix Form

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Today

The Matrix Form and Its Geometry

Partialling Out in General

Tuesday built everything in scalar form. Today: the same objects again in matrix form, for general understanding, followed by lab.

★ = essential, you must understand this

★ = for understanding, not examined

Matrix Form

Why Bother

- With k regressors, scalar algebra needs k normal equations and becomes unreadable. Matrix algebra stays two lines long for any k .
- The **geometry**, projection onto a subspace, is invisible in scalar form and obvious in matrix form.
- In the literature, most estimators you will meet (fixed effects, IV, 2SLS, ridge) are *written* as small variations on $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$. We will keep working in scalars, but recognizing this form is what lets you read modern papers.

But

Nothing in this section is new content. It is the *same* normal equations you derived on Tuesday, written more compactly. The lab that follows practices every piece of linear algebra we use.

Vectors and Matrices, From Zero ★

No linear algebra is assumed in this course. Here is the part we need, from nothing.

- A **vector** is a list of numbers written in a column. X_i is the list describing unit i : one entry per variable, with a 1 on top for the intercept.
- A **matrix** is a rectangular table of numbers: an $n \times k$ matrix has n rows and k columns.
- The **transpose** $'$ tips a column onto its side: if X_i is a $k \times 1$ column, then X_i' is the same list lying down, $1 \times k$.

The one formula to internalize

$$X_i' \beta = \beta_0 \cdot 1 + \beta_1 X_{i1} + \cdots + \beta_{k-1} X_{i,k-1} :$$

a row times a column is a *single number*: multiply matching entries, then add. So $X_i' \beta$ is nothing but unit i 's fitted value, written compactly. Whenever you see $X_i' \beta$, read it as “the regression line evaluated at unit i .”

Multiplying and Inverting

The only two operations we use. **Multiplication** extends the row-times-column rule: entry (r, c) of $\mathbf{AB}$ is (row r of $\mathbf{A}$) times (column c of $\mathbf{B}$):

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}.$$

Dimensions must match, $(n \times k)(k \times 1) = (n \times 1)$, and order matters: $\mathbf{AB} \neq \mathbf{BA}$ in general.

Inversion is the matrix version of dividing. For a number, $a^{-1}a = 1$; for a square matrix, $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$, the *identity* matrix (ones on the diagonal, zeros elsewhere), which leaves whatever it multiplies unchanged.

- Just as $1/a$ fails to exist when $a = 0$, $\mathbf{A}^{-1}$ fails to exist when the matrix collapses information; that is the “rank” condition, explained shortly.
- You will never invert a matrix by hand here; R’s `solve()` does it. What matters is knowing what the symbol says.

Stacking the Data

For n units and k regressors (including the constant):

$$\mathbf{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}_{n \times 1}, \quad \mathbf{X} = \begin{bmatrix} X'_1 \\ \vdots \\ X'_n \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1,k-1} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{n,k-1} \end{bmatrix}_{n \times k}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The model, stacked over units:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad \Longleftrightarrow \quad Y_i = X'_i\boldsymbol{\beta} + \epsilon_i, \quad i = 1, \dots, n.$$

Two objects do all the work: $\mathbf{X}'\mathbf{X} = \sum_i X_i X'_i$ (a $k \times k$ matrix) and $\mathbf{X}'\mathbf{y} = \sum_i X_i Y_i$ (a $k \times 1$ vector).

Least Squares in Matrix Form: the Expansion

The sum of squared errors is the error vector times itself:

$$S(b) = \sum_i (Y_i - X_i' b)^2 = (\mathbf{y} - \mathbf{X}b)'(\mathbf{y} - \mathbf{X}b).$$

Two transpose rules, used constantly

$(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$, and $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$: transposing a product *reverses the order*.
Hence $(\mathbf{X}b)' = b'\mathbf{X}'$.

Multiply out, term by term:

$$S(b) = \mathbf{y}'\mathbf{y} - \mathbf{y}'\mathbf{X}b - b'\mathbf{X}'\mathbf{y} + b'\mathbf{X}'\mathbf{X}b = \mathbf{y}'\mathbf{y} - 2b'\mathbf{X}'\mathbf{y} + b'\mathbf{X}'\mathbf{X}b.$$

- Why do the middle terms merge? Each is 1×1 , a single number, and a number equals its own transpose: $(b'\mathbf{X}'\mathbf{y})' = \mathbf{y}'\mathbf{X}b$. The two terms are the *same* number written twice.

Minimizing: Two Matrix Derivatives, From Zero

$\partial S/\partial b$ just stacks the k partials into a column; two stock rules cover everything:

The rules, with their scalar shadows

$$\frac{\partial(a'b)}{\partial b} = a \quad \text{since } a'b = \sum_j a_j b_j \text{ gives } \partial/\partial b_j = a_j \quad (\text{cf. } \frac{d}{db} ab = a)$$

$$\frac{\partial(b'Ab)}{\partial b} = 2Ab \quad \text{for symmetric } A \quad (\text{cf. } \frac{d}{db} ab^2 = 2ab)$$

Apply them with $a = \mathbf{X}'\mathbf{y}$ and $A = \mathbf{X}'\mathbf{X}$ (symmetric):

$$\frac{\partial S(b)}{\partial b} = -2\mathbf{X}'\mathbf{y} + 2\mathbf{X}'\mathbf{X}b \stackrel{!}{=} 0 \quad \implies \quad \underbrace{\mathbf{X}'\mathbf{X} \hat{\beta}}_{\text{normal equations}} = \mathbf{X}'\mathbf{y}.$$

The *same* k equations you wrote out one at a time on Tuesday. Why a minimum?

$v'\mathbf{X}'\mathbf{X}v = \|\mathbf{X}v\|^2 \geq 0$ for all v : S is a bowl. (Shorthand: $\mathbf{X}'\mathbf{X} \succeq 0$, the matrix $\succeq$.)

The Normal Equations, Slowly: the Bivariate Case

Take $k = 2$ with $X_i = (1, X_i)'$, so $b = (b_0, b_1)'$. The two building blocks are

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Write out $\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{y}$ one row at a time:

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_i X_i = \sum_i Y_i \iff \sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

$$\hat{\beta}_0 \sum_i X_i + \hat{\beta}_1 \sum_i X_i^2 = \sum_i X_i Y_i \iff \sum_i X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0$$

- Row by row, the matrix equation *is* Tuesday's (N1) and (N2): residuals sum to zero, and residuals are orthogonal to the regressor.
- One matrix line, k scalar equations. That is the entire content of the compact notation: nothing new, everything stacked.

The Estimator, Existence, and Rank

What “rank” means, in words

The **rank** of $\mathbf{X}$ is the number of genuinely distinct columns it contains: columns that cannot be rebuilt as combinations of the others. *Full column rank*, $\text{rank}(\mathbf{X}) = k$, says no column is redundant: the data carry k separate pieces of information, one per coefficient.

The OLS estimator

If $\text{rank}(\mathbf{X}) = k$, $\mathbf{X}'\mathbf{X}$ is invertible and

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = (\mathbb{E}_n[X_i X_i'])^{-1} \mathbb{E}_n[X_i Y_i].$$

- Full rank needs $n \geq k$; it is the matrix version of “ X must vary.”
- Failures you will actually meet: the *dummy variable trap*; including a variable and a rescaling of it; a group indicator constant within a fixed effect.
- If rank fails, R silently drops a column and reports NA. That NA is never a mystery.

Sanity Check: Recovering the Scalar Formula ★

Let $k = 2$ with $X_i = (1, X_i)'$. Then

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \quad \mathbf{X}'\mathbf{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Using $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$

$$\hat{\beta}_1 = \frac{n \sum_i X_i Y_i - \sum_i X_i \sum_i Y_i}{n \sum_i X_i^2 - (\sum_i X_i)^2} = \frac{\text{Cov}_n(X, Y)}{\text{Var}_n(X)}, \quad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Exactly what we derived by hand. The matrix form is a change of language, not of substance.

Where Projection Comes From: a Change of Picture ★

Every scatterplot you have drawn puts *variables* on the axes and observations as points: n points in 2 dimensions. Now flip it:

- Let every *observation* be an axis: $\mathbb{R}^n$, one dimension per person. The whole variable $\mathbf{y}$ becomes a *single arrow* in that space; so does each column of $\mathbf{X}$.
- As b ranges over all values, the fitted vectors $\mathbf{X}b$ sweep out a k -dimensional plane through the origin: every fit the model could ever produce.
- Least squares asks for the point of that plane *closest* to $\mathbf{y}$. Geometry answers: drop a perpendicular. Its foot is $\hat{\mathbf{y}}$; the perpendicular segment itself is $\hat{\mathbf{e}}$.
- Perpendicular to the plane = at right angles to every column of $\mathbf{X}$, that is, $\mathbf{X}'\hat{\mathbf{e}} = \mathbf{0}$. **These are the normal equations**, and “normal” is the Latin word for perpendicular. The name was the geometry all along.

“Projection” is simply the operation *drop the perpendicular*. The next frames give it a matrix.

OLS Is a Projection ★

Think of $\mathbf{y}$ as a single point in $\mathbb{R}^n$, one dimension per *observation*. The k columns of $\mathbf{X}$ span a k -dimensional subspace

$$\text{col}(\mathbf{X}) = \{\mathbf{X}b : b \in \mathbb{R}^k\} \subset \mathbb{R}^n.$$

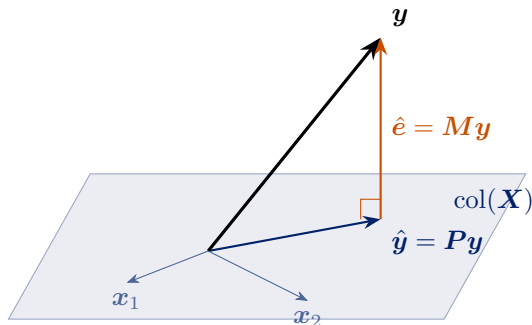
OLS finds the point in $\text{col}(\mathbf{X})$ closest to $\mathbf{y}$: $\hat{\mathbf{y}} = \arg \min_{v \in \text{col}(\mathbf{X})} \|\mathbf{y} - v\|^2$.

Projection and annihilator matrices

$$\mathbf{P} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \text{ ("hat matrix")}, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{P} \text{ ("residual maker")}$$

so that $\hat{\mathbf{y}} = \mathbf{P}\mathbf{y}$ and $\hat{\mathbf{e}} = \mathbf{M}\mathbf{y}$.

The Picture ★



- $\hat{e} \perp \text{col}(\mathbf{X})$, the geometric content of $\mathbf{X}'\hat{e} = \mathbf{0}$, which is the geometric content of “residuals are uncorrelated with the regressors.”
- Fitting is dropping a perpendicular. Everything else is bookkeeping.

Properties of P and M ★

Symmetric

$$P' = P, \quad M' = M$$

Idempotent

$$PP = X(X'X)^{-1} \underbrace{X'X(X'X)^{-1}}_I X' = P$$

$$MM = (I - P)(I - P) = M$$

Mutually orthogonal

$$PM = P - PP = 0$$

For the traces: $\text{tr}(P) = \text{tr}((X'X)^{-1}X'X) = \text{tr}(I_k) = k$, using the cyclic property $\text{tr}(ABC) = \text{tr}(BCA)$, which we use repeatedly.

Act correctly on X

$$PX = X, \quad MX = 0$$

Traces = dimensions

$$\text{tr}(P) = k$$

$$\text{tr}(M) = n - k$$

Eigenvalues are all 0 or 1; P has k ones, M has $n - k$.

Leverage ★

The i th diagonal element of $\mathbf{P}$,

$$h_{ii} = X_i'(\mathbf{X}'\mathbf{X})^{-1}X_i = \frac{\partial \hat{Y}_i}{\partial Y_i},$$

is the **leverage** of observation i : how much unit i 's own outcome pulls its own fitted value.

- $0 \leq h_{ii} \leq 1$ and $\sum_i h_{ii} = k$, so average leverage is k/n .
- High leverage means unusual *covariate* values, not an unusual outcome.
- $\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii})$: high-leverage points have artificially small residuals, they drag the line toward themselves.
- Leverage returns next week in the finite-sample corrections HC2 and HC3.

Sums of Squares, Again ★

Because $\mathbf{y} = \mathbf{P}\mathbf{y} + \mathbf{M}\mathbf{y}$ and $\mathbf{P}\mathbf{M} = \mathbf{0}$, Pythagoras gives

$$\mathbf{y}'\mathbf{y} = \mathbf{y}'\mathbf{P}\mathbf{y} + \mathbf{y}'\mathbf{M}\mathbf{y} = \hat{\mathbf{y}}'\hat{\mathbf{y}} + \hat{\mathbf{e}}'\hat{\mathbf{e}}.$$

Subtracting $n\bar{Y}^2$ (valid when a constant is included) returns the scalar decomposition $TSS = ESS + RSS$.

- R^2 is also $\cos^2$ of the angle between $\mathbf{y} - \bar{Y}\mathbf{1}$ and $\hat{\mathbf{y}} - \bar{Y}\mathbf{1}$: fit is literally about angles.
- Adding a regressor enlarges $\text{col}(\mathbf{X})$, which can only move $\hat{\mathbf{y}}$ closer to $\mathbf{y}$, the geometric reason R^2 never falls.

Unbiasedness, in Matrix Form ★

Tuesday's two-line scalar proof, now for any k :

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \epsilon) = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\epsilon,$$

$$E[\hat{\beta} \mid \mathbf{X}] = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \underbrace{E[\epsilon \mid \mathbf{X}]}_{=0} = \beta. \quad \blacksquare$$

- Note where the work is done: *only* by A4.
- Heteroskedasticity, clustering, and non-normality do **not** bias $\hat{\beta}$. They affect its *variance*, next week.
- A4 fails whenever an omitted variable, reverse causation, measurement error in X , or selection puts something correlated with X into ϵ . That is the entire causal-inference problem in one sentence.

The Gauss–Markov Theorem ★

Theorem

Under A1–A5, $\hat{\beta}$ is the **Best Linear Unbiased Estimator**: among all $\tilde{\beta} = \mathbf{C}\mathbf{y}$ with $\mathbf{C}$ a function of $\mathbf{X}$ and $E[\tilde{\beta} \mid \mathbf{X}] = \beta$, $\text{Var}(\tilde{\beta} \mid \mathbf{X}) - \text{Var}(\hat{\beta} \mid \mathbf{X}) \succeq 0$.

The Gauss–Markov Theorem ★

Theorem

Under A1–A5, $\hat{\beta}$ is the **Best Linear Unbiased Estimator**: among all $\tilde{\beta} = C\mathbf{y}$ with C a function of $\mathbf{X}$ and $E[\tilde{\beta} | \mathbf{X}] = \beta$, $\text{Var}(\tilde{\beta} | \mathbf{X}) - \text{Var}(\hat{\beta} | \mathbf{X}) \succeq 0$.

Proof. Write $C = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}$. Unbiasedness for all β forces $\mathbf{D}\mathbf{X} = \mathbf{0}$. Then

$$\begin{aligned}\text{Var}(C\mathbf{y} | \mathbf{X}) &= \sigma^2 [(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' + \mathbf{D}] [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} + \mathbf{D}'] \\ &= \sigma^2 (\mathbf{X}'\mathbf{X})^{-1} + \sigma^2 \underbrace{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{D}'}_{=0} + \sigma^2 \underbrace{\mathbf{D}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}}_{=0} + \sigma^2 \mathbf{D}\mathbf{D}' \\ &= \text{Var}(\hat{\beta} | \mathbf{X}) + \sigma^2 \mathbf{D}\mathbf{D}' \succeq \text{Var}(\hat{\beta} | \mathbf{X}). \quad \blacksquare\end{aligned}$$

Partialling Out

The Frisch–Waugh–Lovell Theorem ★

Partition $\mathbf{X} = [\mathbf{X}_1 \ \mathbf{X}_2]$: a treatment of interest and controls. Let $\mathbf{M}_2 = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$ and define

$$\tilde{\mathbf{X}}_1 = \mathbf{M}_2\mathbf{X}_1, \quad \tilde{\mathbf{y}} = \mathbf{M}_2\mathbf{y}.$$

Theorem

The OLS estimate $\hat{\beta}_1$ from the full regression equals

$$\hat{\beta}_1 = (\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1'\tilde{\mathbf{X}}_1)^{-1}\tilde{\mathbf{X}}_1'\mathbf{y},$$

and the residuals from the short regression are numerically identical to $\hat{\mathbf{e}}$ from the full one.

This is the three-step recipe from earlier, now for any number of controls.

Proof of FWL ★

Premultiply the model by M_2 and use $M_2X_2 = 0$:

$$M_2y = M_2X_1\beta_1 + \underbrace{M_2X_2}_{=0}\beta_2 + M_2\epsilon \quad \implies \quad \tilde{y} = \tilde{X}_1\beta_1 + M_2\epsilon.$$

Proof of FWL ★

Premultiply the model by M_2 and use $M_2X_2 = 0$:

$$M_2y = M_2X_1\beta_1 + \underbrace{M_2X_2}_{=0}\beta_2 + M_2\epsilon \implies \tilde{y} = \tilde{X}_1\beta_1 + M_2\epsilon.$$

For the estimator, decompose the fitted model $y = X_1\hat{\beta}_1 + X_2\hat{\beta}_2 + \hat{e}$ and premultiply by $\tilde{X}_1' = X_1'M_2$:

$$\tilde{X}_1'y = \tilde{X}_1'\tilde{X}_1\hat{\beta}_1 + X_1'\underbrace{M_2X_2}_{=0}\hat{\beta}_2 + \underbrace{X_1'M_2}_{=0}\hat{e},$$

using $M_2\hat{e} = \hat{e}$ (since $X_2'\hat{e} = 0$) and $X_1'\hat{e} = 0$. Solving gives the result. Using $\tilde{y}$ instead of y changes nothing because $\tilde{X}_1'M_2 = \tilde{X}_1'$. ■

Why FWL Matters More Than It Looks

1. **Interpretation.** “Controlling for W ” = discarding all covariance between X_1 and W . Your estimate lives on what remains.
2. **Computation.** Fixed-effects estimators *are* FWL: regressing on unit dummies is the same as demeaning within unit (Week 10).
3. **Diagnostics.** The added-variable plot is the only two-dimensional picture that faithfully represents a multivariate slope.
4. **Generality.** Nothing in the argument requires $\mathbf{X}_2$ to enter linearly, it only requires that we can compute the part of X_1 that $\mathbf{X}_2$ predicts. That observation is the seed of a large literature, and we return to it briefly in Week 13.

Omitted Variables, in Matrix Form ★

With $\mathbf{X}_1$ included and $\mathbf{X}_2$ omitted,

$$E[\tilde{\beta}_1 \mid \mathbf{X}] = \beta_1 + \underbrace{(\mathbf{X}_1' \mathbf{X}_1)^{-1} \mathbf{X}_1' \mathbf{X}_2}_{\hat{\delta}} \beta_2.$$

- Each column of $\hat{\delta}$ is the coefficient vector from regressing one omitted variable on all the included ones, exactly δ_1 from the scalar case, stacked.
- With several omitted variables the bias is a *sum* of terms which may offset, so “the bias must be positive” arguments require care.
- Sensitivity analysis (Week 6) formalizes the question: how strong would an omitted confounder have to be to overturn the result?

Where This Leaves Us

- OLS minimizes squared errors. Everything else, the slope formula, the normal equations, FWL, OVB, follows from that one criterion by algebra.
- In the population, OLS estimates the best linear approximation to the CEF. Under A1–A4 the sample version is unbiased for it.
- Whether that coefficient is a *causal effect* is a separate question that regression cannot answer.
- **Next week:** we have $\hat{\beta}$; now we need to know how uncertain it is, and the standard error your software prints is usually the wrong one.

This week's reading

MHE Chapter 3; CCI Chapter 6. The lab practices the matrix algebra hands-on and computes OLS by hand in R.

References

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